Anshul Subramanian

AI Engineer | Edge AI and Computer Vision | Generative Models

+91 9999885342 | anshulsubramanianapplications@gmail.com | India

LinkedIn | GitHub | Portfolio

SUMMARY

AI Engineer with 4.5+ years of experience at Samsung Research building production-grade machine learning systems for mobile devices. Specialized in Edge AI, Generative Models, and Computer Vision for smartphone imaging pipelines. Experienced in deploying GAN-based image translation systems, computational photography models, and real-time computer vision pipelines on commercial smartphones. Author of patents in computational photography and mobile AI systems. Skilled in PyTorch, TensorFlow, Android ML integration, and deep learning model optimization for real-time mobile inference.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL

ML Frameworks: PyTorch, TensorFlow, Keras, ONNX, TensorFlow Lite

Native and Mobile: Android NDK, JNI (Java Native Interface), Shared Libraries (.SO), Android ML Integration, C/C++ Android Integration

AI / ML: Deep Learning, Computer Vision, Generative AI, GANs, Diffusion Models, Transformers, Large Language Models (LLMs), Transfer Learning, Self-Supervised Learning

Specializations: Edge AI, On-Device Inference, Optical Flow, Image Segmentation, Object Detection, Image-to-Image Translation, Computational Photography, Neural Architecture Search

Tools and Infra: OpenCV, CUDA, Android ML Integration, Model Quantization, Model Pruning, Model Optimization, Dataset Engineering, MLOps, Git, Linux

PROFESSIONAL EXPERIENCE

Samsung R&D Institute, Noida | AI Engineer

Dec 2021 - Present

Generative AI and Computer Vision

- Designed and deployed a Conditional GAN image-to-image translation network for mobile photography; curated a 115k+ image-pair training dataset and optimized model for on-device inference (under 4 seconds) via quantization and architecture pruning, achieving image quality comparable to diffusion models at approximately 10x smaller model footprint.
- Led end-to-end development of a GAN-based focal length translation system; designed hybrid architecture combining facial-region image translation (GAN) with CNN-based background inpainting and reconstruction; achieved PSNR above 30 dB and LPIPS below 0.1 at under 300 ms inference latency. Filed as patent application (2024).
- Led a 5-engineer team to build a real-time GAN-based lens distortion correction pipeline integrating multi-plane reprojection algorithms; applied model optimization and quantization to achieve LPIPS below 0.1 and sub-300 ms inference, enabling live viewfinder correction on mobile hardware.

Computational Photography

- Developed a deep learning pipeline for long-exposure image synthesis from short video clips; combined optical flow estimation, motion-aware frame interpolation, and temporal aggregation to simulate realistic motion blur entirely on-device without cloud inference.
- Fine-tuned and deployed a mobile facial recognition model integrated into the camera ISP pipeline; applied transfer learning and on-device inference optimization for real-time performance within Samsung's flagship camera stack.

Platform and Architecture

- Integrated production ML models (portrait segmentation, preview, capture) into Android camera framework; improved HDR frame alignment, semantic segmentation accuracy, and temporal consistency across frames.
- Designed a novel attention mechanism for Natural Bokeh Gradation neural network models; architected and trained GAN-based image style transfer systems currently deployed in production on Samsung devices.

- Built an internal Figma-to-Android-XML code generation tool using computer vision and layout parsing, eliminating manual UI implementation work for the mobile engineering team.

Product Impact

- Presented AI camera features at six Samsung flagship device launch events (Galaxy S/Z series).
- Resolved 70+ camera application issues across performance, frame consistency, and stability, directly improving shipped product quality.

Incedo Inc, Gurgaon | Software Development Engineer

Jul 2021 - Nov 2021

- Built Python/SQL automation pipelines for enterprise clients, cutting a daily data extraction and reporting workflow from 2 hours to under 2 minutes (98% reduction).

SELECTED PROJECTS

GAN Based Time-Weather Translation

- Designed and trained a Conditional GAN image-to-image translation network to perform time-of-day and weather condition transformations on mobile photography (e.g. day-to-night, clear-to-rainy).
- Curated a 115k+ image-pair training dataset with diverse lighting and weather conditions; applied data augmentation and dataset engineering techniques to improve generalization across scenes.
- Optimized model for on-device inference via quantization and architecture pruning, achieving under 4 second latency on smartphone hardware with image quality comparable to diffusion models at approximately 10x smaller model footprint.

On-Device AI Agent (NanoAgent)

- Designed and implemented an on-device LLM-powered AI agent capable of executing 30+ smartphone actions without cloud dependency; built modular tool-calling architecture with a lightweight local language model for private, real-time inference on mobile hardware.
- Implemented dual-inference pipeline using Gemini Nano via ML Kit Prompt API with fallback to local models for broader device compatibility; optimized end-to-end latency for on-device natural language understanding and task execution.

Mathematical OCR using Transformers

- Built a Vision Transformer (ViT) encoder-decoder system for end-to-end recognition and parsing of mathematical expressions from images; constructed and augmented a custom 2,400+ sample dataset covering powers, fractions, and multi-operator expressions.

PATENTS

Method and System for Processing Images of an Under Display Camera

Patent Application | 2022

Method and System for Converting Images from Low Focal Length to High Focal Length

Patent Application | 2024

EDUCATION

SRM Institute of Science and Technology | 2017 - 2021

B.Tech, Computer Science | GPA: 91.25%

CERTIFICATIONS

- Applied Deep Learning for Computer Vision, IIT Ropar, 2024
- Introduction to Generative AI, Google, 2023